Pablo Piantanida

Research publications – accompanying the industry research CV

Full Publication List

Preprints and submitted/revised articles

[P1] P. Formont, M. Darrin, I. B. Ayed, and P. Piantanida, "MolRGen: A Training and Evaluation Setting for De Novo Molecular Generation with Reasonning Models," 2026. [paper] [arXiv]

[P2] P. Piantanida and G. Pichler, "A theory for rational speech acts as KL projections with powered semantics," 2026, (submitted)

[P3] Y. Bahram, Z. Dehghani, M. Desbos, E. Granger, P. Piantanida, and M. Shateri, "Continuous adversarial meanflow transfer," 2026. [paper] [arXiv]

[P4] C. Chataigner, P. Piantanida, and G. Farnadi, "Position: Auditing Is Not Evaluating; LLM Audit Requires Dynamic, Contextual, Budget-Aware and Reliable Evidence," Feb. 2026, working paper or preprint. [paper]

[P5] L. Estienne, E. Ernst, M. Vera, P. Piantanida, and L. Ferrer, "Ecuas$_n$: A family of metrics for principled evaluation of uncertainty-augmented systems," 2026. [paper] [arXiv]

Proceedings of international conferences

[C1] A. Satouf, G. D'Erasmo, Y. Zong, H. Amadou-Boubacar, P. Piantanida, and B. Piwowarski, "STORM: Stepwise token optimization with reward-guided beam search," in *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, 2026. [paper] [arXiv]

[C2] L. Fosse, P. Formont, E. Aubinais, and P. Piantanida, "Data-driven estimation of statistical deficiency," in *2026 IEEE Information Theory Workshop (ITW)*, 2026

[C3] E. Aubinais, P. Formont, P. Piantanida, and E. Gassiat, "Membership Inference Risks in Quantized Models: A Theoretical and Empirical Study," in *The 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026. [paper] [arXiv]

[C4] G. Pichler, M. Romanelli, and P. Piantanida, "Happiness as a Measure of Fairness," in *The 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026. [paper] [arXiv]

[C5] A. Satouf, Y. Zong, H. A. Boubacar, P. Piantanida, and B. Piwowarski, "QueStER: Query Specification for Generative Keyword-Based Retrieval," in *Findings of the Association for Computational Linguistics: EACL 2026*, ser. Findings of ACL, V. Demberg, K. Inui, and L. Marquez, Eds. Association for Computational Linguistics, March 2026, pp. 5957–5968. [paper] [arXiv]

[C6] A. Satouf, G. B. Zenou, B. Piwowarski, H. A. Boubacar, and P. Piantanida, "Rational Retrieval Acts: Leveraging Pragmatic Reasoning to Improve Sparse Retrieval," in *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, N. Ferro, M. Maistro, G. Pasi, O. Alonso, A. Trotman, and S. Verberne, Eds. ACM, 2025, pp. 2674–2678. [paper] [arXiv]

[C7] P. Marza, L. Fillioux, S. Boutaj, K. Mahatha, C. Desrosiers, P. Piantanida, J. Dolz, S. Christodoulidis, and M. Vakalopoulou, "THUNDER: Tile-level Histopathology image UNDERstanding benchmark," in *Advances in Neural Information Processing Systems 38*. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2025, pp. 171 942–172 034, (spotlight). [paper] [arXiv]

[C8] P. Formont, M. DARRIN, B. Karimian, E. Granger, J. C. Cheung, I. B. Ayed, M. Shateri, and P. Piantanida, "Learning Task-Agnostic Representations through Multi-Teacher Distillation," in *Advances in Neural Information Processing Systems 38*. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2025, pp. 121 561–121 607. [paper] [arXiv]

[C9] C. Spinoso-Di Piano, D. E. Austin, P. Piantanida, and J. C. Cheung, "(RSA)^2: A Rhetorical-Strategy-Aware Rational Speech Act Framework for Figurative Language Understanding," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds. Association for Computational Linguistics, July 2025, pp. 20 898–20 938. [paper] [arXiv]

[C10] L. Estienne, G. B. Zenou, N. Naderi, J. C. Cheung, and P. Piantanida, "Collaborative Rational Speech Act: Pragmatic Reasoning for Multi-Turn Dialog," in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, November 2025, pp. 22 509–22 523, (oral). [paper] [arXiv]

[C11] L. Fosse, F. Bechet, B. Favre, G. Damnati, G. Lecorvé, M. Darrin, P. Formont, and P. Piantanida, "Statistical Deficiency for Task Inclusion Estimation," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds. Association for Computational Linguistics, July 2025, pp. 382–415. [paper] [arXiv]

[C12] M. Dubois, F. Yvon, and P. Piantanida, "MOSAIC: Multiple Observers Spotting AI Content," in *Findings of the Association for Computational Linguistics: ACL 2025*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds. Association for Computational Linguistics, July 2025, pp. 24 230–24 247. [paper] [arXiv]

[C13] ——, "MOSAIC at GENAI Detection Task 3 : Zero-Shot Detection Using an Ensemble of Models," in *Proceedings of the 1stWorkshop on GenAI Content Detection (GenAIDetect)*. International Conference on Computational Linguistics, January 2025, pp. 371–376. [paper]

[C14] O. Beaumont, R. Bourgouin, M. Darrin, L. Marchal, and P. Piantanida, "Leveraging Expert Usage to Speed up LLM Inference with Expert Parallelism," in *Euro-Par 2025: Parallel Processing - 31st European Conference on Parallel and Distributed Processing, Dresden, Germany, August 25-29, 2025, Proceedings, Part I*, ser. Lecture Notes in Computer Science, W. E. Nagel, D. Goehringer, and P. C. Diniz, Eds., vol. 15900. Springer, 2025, pp. 145–158. [paper]

[C15] M. Dubois, F. Yvon, and P. Piantanida, "How Sampling Affects the Detectability of Machine-written texts: A Comprehensive Study," in *Findings of the Association for Computational Linguistics: EMNLP 2025*, C. Christodoulopoulos, T. Chakraborty, C. Rose, and V. Peng, Eds. Suzhou, China: Association for Computational Linguistics, November 2025, pp. 11 369–11 387. [paper] [arXiv]

[C16] C. Presvôts, M. Kieffer, T. Prevost, P. Panciatici, Z. Li, and P. Piantanida, "Two-stage Multiple-Model Compression Approach for Sampled Electrical Signals," in *2024 Data Compression Conference (DCC)*, A. Bilgin, J. E. Fowler, J. Serra-Sagristà, Y. Ye, and J. A. Storer, Eds. IEEE, 2024, pp. 522–531. [paper]

[C17] M. Darrin, P. Formont, I. B. Ayed, J. C. Cheung, and P. Piantanida, "When is an Embedding Model More Promising than Another?" in *Advances in Neural Information Processing Systems 37*. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2024, pp. 68 330–68 379. [paper] [arXiv]

[C18] A. Altieri, M. Romanelli, G. Pichler, F. Alberge, and P. Piantanida, "Beyond the Norms: Detecting Prediction Errors in Regression Models," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 21–27 Jul 2024, pp. 1186–1221, (spotlight). [paper] [arXiv]

[C19] E. D. C. Gomes, M. Romanelli, G. Pichler, and P. Piantanida, "A Data-Driven Measure of Relative Uncertainty for Misclassification Detection," in *The Twelfth International Conference on Learning Representations*, 2024. [paper] [arXiv]

[C20] M. Darrin, P. Formont, J. Cheung, and P. Piantanida, "COSMIC: Mutual Information for Task-Agnostic Summarization Evaluation," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, August 2024, pp. 12 696–12 717, (oral). [paper] [arXiv]

[C21] M. Darrin, I. Arous, P. Piantanida, and J. Cheung, "GLIMPSE: Pragmatically Informative Multi-Document Summarization for Scholarly Reviews," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, August 2024, pp. 12 737–12 752. [paper] [arXiv]

[C22] F. Granese, M. Romanelli, and P. Piantanida, "Optimal Zero-Shot Detector for Multi-Armed Attacks," in *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, S. Dasgupta, S. Mandt, and Y. Li, Eds., vol. 238. PMLR, 02–04 May 2024, pp. 2467–2475. [paper] [arXiv]

[C23] D. Gorla, L. Jalouzot, F. Granese, C. Palamidessi, and P. Piantanida, "On the (Im)Possibility of Estimating Various Notions of Differential Privacy (short paper)," in *Proceedings of the 24th Italian Conference on Theoretical Computer Science, Palermo, Italy, September 13-15, 2023*, ser. CEUR Workshop Proceedings, G. Castiglione and M. Sciortino, Eds., vol. 3587. CEUR-WS.org, 2023, pp. 219–224. [paper] [arXiv]

[C24] P. Colombo, M. Picot, N. Noiry, G. Staerman, and P. Piantanida, "Toward Stronger Textual Attack Detectors," in *Findings of the Association for Computational Linguistics: EMNLP 2023*, H. Bouamor, J. Pino, and K. Bali, Eds. Association for Computational Linguistics, 2023, pp. 484–505. [paper] [arXiv]

[C25] M. Darrin, G. Staerman, E. D. C. Gomes, J. C. Cheung, P. Piantanida, and P. Colombo, "Unsupervised Layer-wise Score Aggregation for Textual OOD Detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38. Association for the Advancement of Artificial Intelligence (AAAI), February 2024, pp. 17 880–17 888. [paper] [arXiv]

[C26] P. Colombo, M. Peyrard, N. Noiry, R. West, and P. Piantanida, "The Glass Ceiling of Automatic Evaluation in Natural Language Generation," in *Findings of the Association for Computational Linguistics: IJCNLP-AACL 2023 (Findings)*, J. C. Park, Y. Arase, B. Hu, W. Lu, D. Wijaya, A. Purwarianti, and A. A. Krisnadhi, Eds. Nusa Dua, Bali: Association for Computational Linguistics, November 2023, pp. 178–183. [paper] [arXiv]

[C27] M. Darrin, P. Piantanida, and P. Colombo, "RainProof: An Umbrella to Shield Text Generator from Out-Of-Distribution Data," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, H. Bouamor, J. Pino, and K. Bali, Eds. Singapore: Association for Computational Linguistics, December 2023, pp. 5831–5857. [paper] [arXiv]

[C28] P. Colombo, N. Noiry, G. Staerman, and P. Piantanida, "A Novel Information Theoretic Objective to Disentangle Representations for Fair Classification," in *Findings of the Association for Computational Linguistics: IJCNLP-AACL 2023 (Findings)*, J. C. Park, Y. Arase, B. Hu, W. Lu, D. Wijaya, A. Purwarianti, and A. A. Krisnadhi, Eds. Nusa Dua, Bali: Association for Computational Linguistics, November 2023, pp. 184–198. [paper] [arXiv]

[C29] P. Colombo, V. Pellegrain, M. Boudiaf, M. Tami, V. Storchan, I. Ayed, and P. Piantanida, "Transductive Learning for Textual Few-Shot Classification in API-based Embedding Models," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, H. Bouamor, J. Pino, and K. Bali, Eds. Singapore: Association for Computational Linguistics, December 2023, pp. 4214–4231. [paper] [arXiv]

[C30] M. Boudiaf, E. Bennequin, M. Tami, C. Hudelot, A. Toubhans, P. Piantanida, and I. B. Ayed, "Open-Set Likelihood Maximization for Few-Shot Learning," in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2023, pp. 24 007–24 016. [paper] [arXiv]

[C31] P. Colombo, E. Dadalto, G. Staerman, N. Noiry, and P. Piantanida, "Beyond Mahalanobis Distance for Textual OOD Detection," in *Advances in Neural Information Processing Systems 35*, S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, Eds., vol. 35. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2022, pp. 17 744–17 759. [paper] [arXiv]

[C32] N. M. Guerreiro, P. Colombo, P. Piantanida, and A. Martins, "Optimal Transport for Unsupervised Hallucination Detection in Neural Machine Translation," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Toronto, Canada: Association for Computational Linguistics, July 2023, pp. 13 766–13 784. [paper] [arXiv]

[C33] F. Granese, M. Picot, M. Romanelli, F. Messina, and P. Piantanida, "MEAD: A Multi-Armed Approach for Evaluation of Adversarial Examples Detectors," in *Machine Learning and Knowledge Discovery in Databases - European Conference, ECML PKDD 2022, Grenoble, France, September 19-23, 2022, Proceedings, Part III*, ser. Lecture Notes in Computer Science, M. Amini, S. Canu, A. Fischer, T. Guns, P. K. Novak, and G. Tsoumakas, Eds., vol. 13715. Springer Nature Switzerland, 2022, pp. 286–303. [paper] [arXiv]

[C34] G. Pichler, P. J. A. Colombo, M. Boudiaf, G. Koliander, and P. Piantanida, "A Differential Entropy Estimator for Training Neural Networks," in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162. PMLR, 17–23 Jul 2022, pp. 17 691–17 715. [paper] [arXiv]

[C35] P. Colombo, G. Staerman, N. Noiry, and P. Piantanida, "Learning Disentangled Textual Representations via Statistical Measures of Similarity," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 2614–2630, (spotlight). [paper] [arXiv]

[C36] G. Del Grosso, H. Jalalzai, G. Pichler, C. Palamidessi, and P. Piantanida, "Leveraging Adversarial Examples to Quantify Membership Information Leakage," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, Jun 19, 2022 - Jun 24, 2022, New Orleans, Louisiana, Nov 16 2022, pp. 10 389–10 399. [paper] [arXiv]

[C37] S. Espinosa, J. F. Silva, and P. Piantanida, "A Data-Driven Quantization Design for Distributed Testing Against Independence with Communication Constraints," in *ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 22-27 May 2022 in Singapore. 2022, pp. 5238–5242. [paper]

[C38] E. Dadalto Carama Gome, F. Alberge, P. Duhamel, and P. Piantanida, "Igeood: An Information Geometry Approach to Out-of-Distribution Detection ," in *The Tenth International Conference on Learning Representations*, (Virtual) Mon Apr 25th through Fri the 29th 2022. [paper] [arXiv]

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[C41] M. Picot, F. Messina, F. Labeau, and P. Piantanida, "Robust Autoencoder-based State Estimation in Power Systems," in *IEEE ISGT NA 2022: "Technology Solutions for an Evolving Grid"*, February 21 - 24, Washington, DC, USA 2022

[C42] G. Del Grosso, G. Pichler, C. Palamidessi, and P. Piantanida, "Formalizing Attribute and Membership Inference Attacks on Machine Learning Models," in *Privacy Preserving Machine Learning*, Nov. 19th, Virtual ACM CCS 2021 Workshop 2021. [paper]

[C43] E. Dadalto Carama Gome, F. Alberge, P. Duhamel, and P. Piantanida, "Igeood: An Information Geometry Approach to Out-of-Distribution Detection ," in *NeurIPS 2021 Workshop on Distribution Shifts*, Dec 13th, 2021, Virtual 2021

[C44] P. Colombo, G. Staerman, C. Clavel, and P. Piantanida, "Automatic Text Evaluation through the Lens of Wasserstein Barycenters," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, November 2021, pp. 10 450–10 466. [paper] [arXiv]

[C45] F. Granese, M. Romanelli, D. Gorla, C. Palamidessi, and P. Piantanida, "DOCTOR: A Simple Method for Detecting Misclassification Errors," in *Advances in Neural Information Processing Systems*, Dec 6th - 14th, 2021, Virtual 2021, (spotlight). [paper] [arXiv]

[C46] O. Veilleux, M. Boudiaf, P. Piantanida, and I. Ben Ayed, "Realistic evaluation of transductive few-shot learning," in *Advances in Neural Information Processing Systems*, Dec 6th - 14th, 2021, Virtual 2021. [paper] [arXiv]

[C47] M. Benammar, E. Dadalto Carama Gome, and P. Piantanida, "CSI-aided Robust Neural-based Decoders," in *The 11th International Symposium on Topics in Coding*, Virtual August 30th to September 3rd 2021

[C48] P. Colombo, P. Piantanida, and C. Clavel, "A Novel Estimator of Mutual Information for Learning to Disentangle Textual Representations," in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. Association for Computational Linguistics, August 2021, pp. 6539–6550, (spotlight). [paper] [arXiv]

[C49] M. Boudiaf, H. Kervadec, I. M. Ziko, P. Piantanida, I. Ben Ayed, and J. Dolz, "Few-Shot Segmentation Without Meta-Learning: A Good Transductive Inference Is All You Need?" in *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, Virtual Sat, Jun 19, 2021 – Fri, Jun 25, 2021 2021, pp. 13 974–13 983. [paper] [arXiv]

[C50] M. Vera, L. Rey Vega, and P. Piantanida, "Information and Regularization in Restricted Boltzmann Machines," in *ICASSP 2021 - 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, Sun, Jun 6, 2021 – Fri, Jun 11 2021, pp. 3155–3159. [paper]

[C51] M. Shateri, F. Messina, P. Piantanida, and F. Labeau, "Learning Sparse Privacy-Preserving Representations for Smart Meters Data," in *EEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (IEEE SmartGridComm 2021)*, Oct. 25 - Oct. 28 2021. [paper] [arXiv]

[C52] M. Boudiaf, J. Rony, I. M. Ziko, E. Granger, M. Pedersoli, P. Piantanida, and I. Ben Ayed, "A Unifying Mutual Information View of Metric Learning: Cross-Entropy vs. Pairwise Losses," in *European Conference on Computer Vision (ECCV)*, 2020, pp. 548–564, (spotlight). [paper] [arXiv]

[C53] M. Boudiaf, I. Ziko, J. Rony, J. Dolz, I. Ben Ayed, and P. Piantanida, "Information Maximization for Few-Shot Learning," in *Advances in Neural Information Processing Systems*, H. Larochelle, M. Ranzato, R. Hadsell, M. F. Balcan, and H. Lin, Eds., vol. 33. Curran Associates, Inc., Dec 5th - 12th, 2020, Vancouver 2020, pp. 2445–2457. [paper] [arXiv]

[C54] M. Romanelli, K. Chatzikokolakis, C. Palamidessi, and P. Piantanida, "Estimating G-Leakage via Machine Learning," in *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security*, ser. CCS 2020. New York, NY, USA: ACM, 2020, pp. 697–716. [paper] [arXiv]

[C55] G. Del Grosso, G. Pichler, and P. Piantanida, "Privacy-Preserving Synthetic Smart Meters Data," in *2021 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT)*. IEEE, February 15-18, Washington, DC, USA 2021, pp. 1–5. [paper] [arXiv]

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[C57] G. Pichler, J. Dolz, I. Ben Ayed, and P. Piantanida, "On Direct Distribution Matching for Adapting Segmentation Networks," in *Proceedings of the Third Conference on Medical Imaging with Deep Learning*, ser. Proceedings of Machine Learning Research, vol. 121. PMLR, 06–08 Jul 2020, pp. 624–637. [paper] [arXiv]

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[C59] C. Feutry, P. Piantanida, and P. Duhamel, "Learning Semi-supervised Anonymized Representations by Mutual Information," in *ICASSP 2020 - 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, May 4 and May 8, Barcelona, Spain 2020, pp. 3467–3471. [paper]

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[C64] E. de Chérisey, S. Guilley, O. Rioul, and P. Piantanida, "An Information-Theoretic Model for Side-Channel Attacks in Embedded Hardware," in *2019 IEEE International Symposium on Information Theory (ISIT)*. IEEE, July 7-12 2019, pp. 310–315. [paper]

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[C74] G. Bassi, M. Skoglund, and P. Piantanida, "Lossy Communication Subject to Statistical Parameter Privacy," in *2018 IEEE International Symposium on Information Theory (ISIT)*. IEEE, June 2018, pp. 1031–1035. [paper]

[C75] A. Bunin, Z. Goldfeld, H. H. Permuter, S. Shamai, P. Cuff, and P. Piantanida, "Key-Message Security over State-Dependent Wiretap Channels," in *2018 IEEE International Symposium on Information Theory (ISIT)*. IEEE, June 2018, pp. 136–140. [paper]

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